

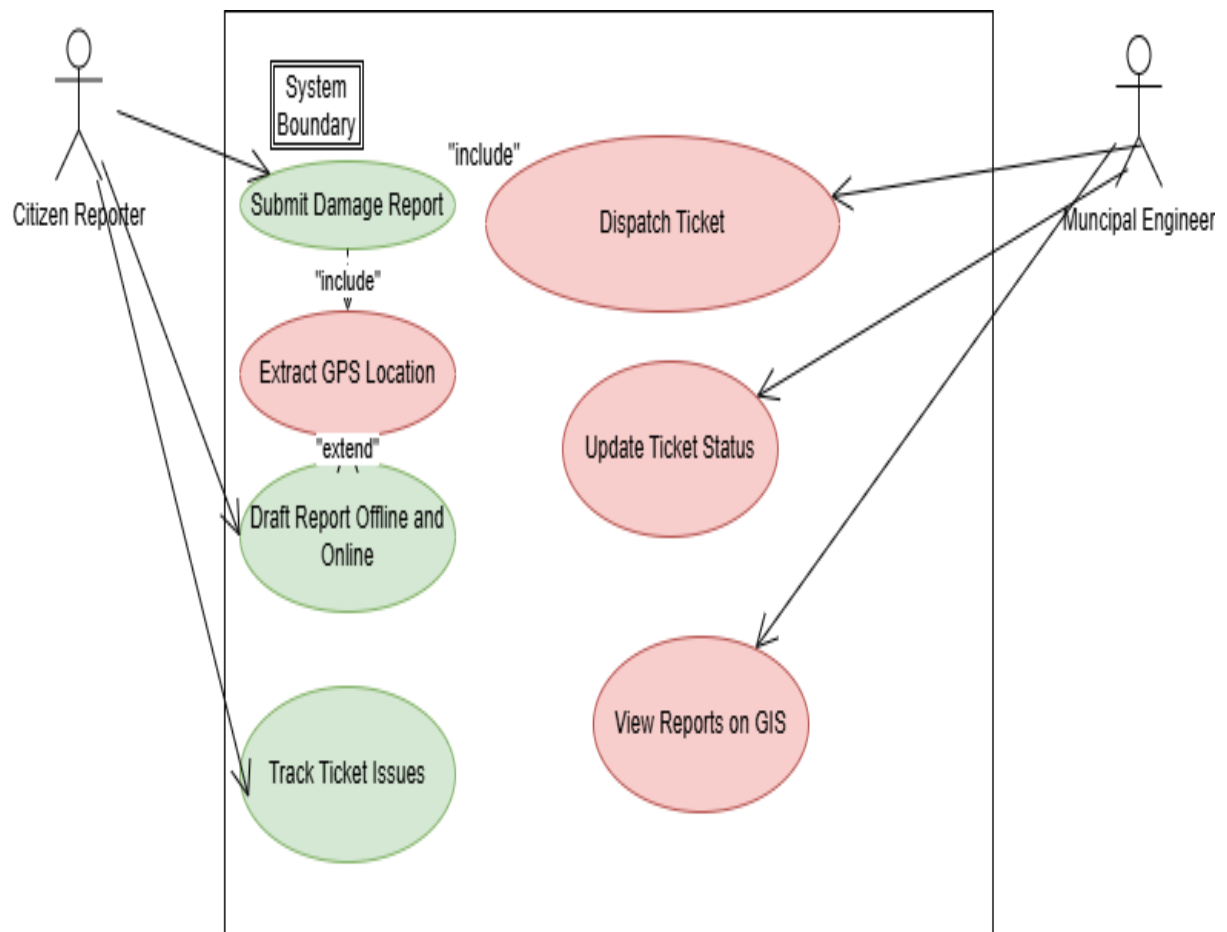
29. Municipal Infrastructure Damage Reporting App
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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint the incident location on the municipal GIS map.	High	Pass: Pothole report is routed to the Road Infrastructure division with exact GPS coordinates attached. Fail: Report is submitted without location data.	Location accuracy is essential for correct department dispatch and field verification.
FR-002	Functional	The system shall allow a Citizen Reporter to capture a new photo or upload an existing photo of the infrastructure damage directly within the app.	High	Pass: A photo thumbnail preview is displayed and attached to the report before submission. Fail: Report is accepted without any photo evidence.	Photographic evidence is required to verify the reported damage and prioritize repairs.
FR-003	Functional	The system shall automatically classify the reported issue by category (Pothole, Streetlight, Water Leak) and dispatch a ticket to the corresponding municipal engineering department (Road Infrastructure, Electrical, or Water) within 5 minutes of submission.	High	Pass: 95% of tickets sampled in testing are auto-routed to the correct department within 5 minutes. Fail: Ticket remains unassigned or is routed to the wrong department after 5 minutes.	Automated routing removes manual triage delay and ensures reports reach the right team quickly.
FR-004	Functional	The system shall allow a Citizen Reporter to view the real-time status of a submitted report (Submitted, Assigned, In Progress, Resolved) using a unique ticket ID.	Medium	Pass: Status shown in the app matches the backend ticket status within 5 minutes of any update. Fail: Displayed status is more than 24 hours out of date.	Status visibility builds citizen trust and reduces duplicate reports and status-check phone calls.

FR-005	Functional	The system shall allow a Municipal Engineer to update a ticket's status and attach resolution notes or a completion photo, which triggers an automatic notification to the reporting citizen.	High	Pass: Citizen receives an in-app or SMS notification within 2 minutes of the engineer marking the ticket Resolved. Fail: No notification is sent, or it arrives more than 30 minutes late.	Closing the feedback loop confirms accountability and lets citizens verify the repair was completed.
NFR-001	Nonfunctional: Performance & Security	The reporting portal shall support offline report drafting and automatically sync the draft (photo, GPS tag, description) once mobile network connectivity resumes.	High	Pass: Benchmark tests confirm drafts sync within 30 seconds of reconnection and meet target latency and security standards under simulated peak load. Fail: Draft data is lost, or sync exceeds 30 seconds after reconnection.	Field-reported issues (e.g., in basements or remote areas) often occur in low- or no-connectivity zones.
NFR-002	Nonfunctional: Security	The system shall encrypt all uploaded photos and GPS location data both in transit and at rest, using TLS 1.2+ for transmission and AES-256 for storage.	High	Pass: Independent security audit confirms TLS 1.2+ is enforced on all API calls and AES-256 is applied to stored media/location data. Fail: Any unencrypted transmission or storage of citizen data is detected.	GPS coordinates and photos are personally identifiable location data and must be protected against interception or breach.

2. UML



3. Use Case Flow Specification

Municipal Infrastructure Damage Reporting App

Use Case: Submit Damage Report

Primary Actor: Citizen Reporter

Supporting Actors: Municipal Engineer, GIS System

Preconditions:

- The Citizen Reporter has the app installed and is logged in.
- The device's camera and location (GPS) permissions are enabled.

Postconditions:

- A damage report is created and saved in the system with a unique ticket ID.
- The ticket is automatically dispatched to the relevant municipal department.
- The Citizen Reporter can track the report under "Track Ticket Issues."

Main Success Scenario

1. Citizen Reporter opens the app and selects "Submit Damage Report."
2. System displays fields to select damage type and attach a photo.
3. Citizen Reporter captures/uploads a photo of the damage.
4. System extracts the GPS location from the photo (*includes "Extract GPS Location"*).
5. System displays the detected location on the map for confirmation.
6. Citizen Reporter confirms the location and submits the report.
7. System validates that all required fields are filled.
8. System dispatches a ticket to the relevant municipal department for action.
9. System displays a confirmation message with the ticket ID.
10. Use case ends successfully.

Alternate Flow-3a:

- 3a1. System detects no active network connection.
- 3a2. System saves the report as a local draft, including the photo and any captured GPS data.
- 3a3. Citizen Reporter is notified the report is saved offline and will submit automatically.
- 3a4. Once connectivity resumes, the system auto-syncs and submits the draft.
- 3a5. Flow resumes at main step 7.

